



Marine IMU Wave-State Estimation with OU-Driven Quaternion MEKF

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Chapter 1. Abstract

This document replaces the earlier trochoidal-wave and raw double-integration draft with the more mature estimator now developed in the Ocean IMU project. The current method is an OU-driven quaternion multiplicative extended Kalman filter, or Q-MEKF, for marine inertial sensing and wave-state estimation. It estimates attitude, heave, 3D displacement, dominant wave frequency, and horizontal wave direction from a calibrated IMU, with optional magnetometer yaw correction and vessel-specific corrections such as IMU lever arm and steady wind heel.

The central change is conceptual. The old draft tried to infer wave height from simplified trochoidal formulas and then use those formulas to help a vertical Kalman integration scheme. The current method instead models the unknown world-frame wave acceleration as a stationary Ornstein-Uhlenbeck, or OU, process and lets the Kalman filter jointly estimate attitude and wave kinematics. Drift is controlled by a soft pseudo-measurement on the integral of displacement, with an adaptive covariance that scales with the sea state.

This formulation is intended for embedded, low-cost marine IMU deployments. It avoids reliance on barometric pressure for wave elevation, avoids uncontrolled double integration of acceleration, and uses a real-time adaptation loop so that the same filter can remain stable across calm chop, long swell, and larger sea states.

Chapter 2. Source and Status

This document is a practical AsciiDoc distillation of the more detailed Ocean IMU mathematical draft:

- [bareboat-necessities/ocean-imu](#)
- [doc/kalman_ou_iii/kalman_ou-w3d.tex](#)
- [doc/kalman_ou_iii/w3d-*.tex-part](#)

It is not meant to duplicate every proof, coefficient table, or plotting fixture from Ocean IMU. Instead, it records the model, state layout, update equations, tuning law, and validation results at a level appropriate for the Bareboat Necessities documentation.

Chapter 3. Motivation

A boat, buoy, or floating sensor experiences wave-induced translational motion, vessel attitude motion, sensor bias, vibration, magnetic disturbance, and measurement noise at the same time. Estimating wave height by simply rotating the accelerometer into a vertical axis and double-integrating it is not stable in practice:

- accelerometer bias integrates into velocity drift and then displacement drift;
- attitude error leaks gravity into the estimated acceleration;
- the initial vertical velocity and displacement are generally unknown;
- low-frequency drift is easily mistaken for long-period swell;
- a barometer does not directly solve the problem on a moving wave surface.

The mature estimator treats the problem as a coupled inertial-navigation and stochastic wave-kinematics problem. The wave acceleration is not assumed to be a perfect sinusoid or trochoid. Instead, it is treated as colored, finite-variance excitation with a correlation time tied to the dominant wave period.

Chapter 4. Estimator Overview

The estimator combines four pieces:

1. a quaternion left-multiplicative EKF for attitude and gyro bias;
2. a 3D linear kinematic chain for acceleration, velocity, displacement, and the integral of displacement;
3. an OU process prior for latent world-frame acceleration;
4. an online sea-state tuner that adapts the OU time constant, acceleration variance, and drift-control pseudo-measurement.

At each IMU step:

- the gyroscope propagates the nominal quaternion;
- the accelerometer contributes a rank-3 EKF measurement of body-frame specific force;
- the magnetometer, when enabled and stable, contributes a rank-3 yaw/field update;
- the pseudo-measurement $\mathbf{S} = \mathbf{0}$ softly constrains long-term displacement drift;
- the frequency tracker and tuner update the sea-state parameters.

Chapter 5. Frames and Signals

The working world frame is NED: north, east, down. The body frame is the calibrated IMU or vessel body frame used by the implementation.

The estimator assumes calibrated triads:

- gyroscope body rate ω_m ;
- accelerometer specific force f_b ;
- magnetometer field m_b , when available.

The gravity vector in NED is

$$g_w = [0, 0, g]^T, \quad g = 9.80665 \text{ m/s}^2.$$

The rotation R_{wb} maps world-frame vectors into the body frame. The nominal attitude is stored as a unit quaternion representing world-to-body orientation.

Chapter 6. State Vector

The extended error-state vector is

$$\mathbf{x} = [\delta\theta, \mathbf{b}_g, \mathbf{v}, \mathbf{p}, S, \mathbf{a}_w, \mathbf{b}_a]^T.$$

The state components are:

Symbol	Meaning
$\delta\theta$	Small attitude-error vector used by the left-multiplicative EKF correction.
$\mathbf{b}_g$	Gyroscope bias, modeled as a random walk.
$\mathbf{v}$	World-frame velocity.
$\mathbf{p}$	World-frame displacement or position relative to the local wave-tracking origin.
S	Integral of displacement, $S(t) = \int_0^t \mathbf{p}(\lambda) d\lambda$. This auxiliary state is used for drift suppression.
$\mathbf{a}_w$	Latent world-frame wave acceleration, modeled as an OU process.
$\mathbf{b}_a$	Accelerometer bias, modeled as a random walk with optional temperature-dependent drift in the measurement model.

With all components enabled, the filter has

$$N_X = 3 + 3 + 3 + 3 + 3 + 3 + 3 = 21$$

states.

Chapter 7. Measurement Model

7.1. Accelerometer

The accelerometer measures body-frame specific force:

$$\mathbf{f}_b = \mathbf{R}_{wb} (\mathbf{a}_w - \mathbf{g}_w) + \mathbf{b}_a(T) + \mathbf{n}_a.$$

Here

$$\mathbf{b}_a(T) = \mathbf{b}_{a0} + \mathbf{k}_a (T - T_{\text{ref}})$$

allows a temperature-dependent accelerometer bias correction, and $\mathbf{n}_a$ is accelerometer measurement noise.

For the left-multiplicative convention, the linearized accelerometer Jacobians are

$$\begin{aligned} d \mathbf{f}_b / d \boldsymbol{\theta} &= -\text{skew}(\mathbf{f}_{\text{cog}}, \mathbf{b}), \\ d \mathbf{f}_b / d \mathbf{a}_w &= \mathbf{R}_{wb}, \\ d \mathbf{f}_b / d \mathbf{b}_a &= \mathbf{I}_3. \end{aligned}$$

The important point is that the accelerometer update does not merely tilt-correct the attitude. Because $d \mathbf{f}_b / d \mathbf{a}_w = \mathbf{R}_{wb}$, it also corrects the latent world-frame acceleration state. That is what ties attitude, gravity alignment, and wave acceleration into one estimator.

7.2. Magnetometer

When magnetometer updates are enabled, the predicted magnetic field is

$$\mathbf{m}_b = \mathbf{R}_{wb} \mathbf{B}_w + \mathbf{n}_m.$$

The world magnetic reference $\mathbf{B}_w$ is obtained from calibration and local magnetic inclination/declination. The attitude Jacobian is

$$d \mathbf{m}_b / d \boldsymbol{\theta} = -\text{skew}(\mathbf{m}_b).$$

In practice, magnetometer yaw correction should be delayed until the tilt estimate is stable and a

reliable local magnetic reference has been accumulated.

7.3. Integral Pseudo-Measurement

To suppress long-term double-integration drift, the filter applies a zero-valued pseudo-measurement to the integral state:

$$\begin{aligned} z_S &= 0 = S + n_S, \\ R_S &= \text{diag}(\sigma_{Sx}^2, \sigma_{Sy}^2, \sigma_{Sz}^2), \\ d z_S / d S &= I_3. \end{aligned}$$

This is not a physical sensor. It is a soft regularizer. Repeated application makes the triple integral behave like a leaky integral: low-frequency drift is attenuated while wave-band motion is still allowed.

The choice of R_S is critical:

- small R_S means strong drift control but risks over-regularizing long swell;
- large R_S means weak drift control and more low-frequency wandering;
- adaptive R_S lets the filter scale this tradeoff with the sea state.

7.4. IMU Lever Arm

If the IMU is not mounted at the vessel center of gravity, angular motion adds apparent acceleration. For a body-frame lever arm r_b , body angular velocity ω_b , and angular acceleration α_b , the accelerometer model becomes

$$f_b = f_{cog,b} + \alpha_b \times r_b + \omega_b \times (\omega_b \times r_b) + b_a(T) + n_a.$$

The lever-arm term is treated as a known kinematic correction. Setting $r_b = 0$ recovers the center-of-gravity model.

Chapter 8. Continuous-Time Process Model

8.1. Quaternion Propagation

The nominal quaternion is propagated from the bias-corrected gyroscope:

$$d q / d t = -1/2 \Omega(\omega_m - b_g) q.$$

Unit norm is enforced in implementation. The gyro bias follows a random walk:

$$d b_g / d t = w_{bg}.$$

The small left-multiplicative attitude error evolves approximately as

$$d(\delta \theta) / d t = -\text{skew}(\omega_m - b_g) \delta \theta + \delta b_g - n_g.$$

8.2. OU-Driven Kinematic Chain

For each world axis, the linear chain is

$$d v / d t = a_w,$$

$$d p / d t = v,$$

$$d S / d t = p.$$

The latent acceleration is modeled by an OU process:

$$d a_w / d t = -(1/\tau) a_w + \sqrt{2/\tau} L_a w_a,$$

$$L_a L_a^T = \Sigma_{aw}.$$

Here τ is the acceleration correlation time and Σ_{aw} is the stationary acceleration covariance. The OU model is not claiming that the ocean surface is literally an OU process. It is a compact Gauss-Markov prior for colored wave excitation with bounded variance and a finite correlation time. That prior is useful because it is Markov, embedded-friendly, and admits analytic discretization.

The accelerometer bias is modeled as

$$d b_a / d t = w_{ba}.$$

Its temperature correction belongs in the measurement model, not in the process model.

8.3. Unified Error-State Form

The continuous-time linearized error dynamics can be written as

$$d x_{err} / d t = F x_{err} + L w,$$

$$P_{dot} = F P + P F^T + Q_c.$$

With measurements, the continuous Kalman-Bucy information term is

$$P_{dot} = F P + P F^T + Q_c - P H^T R^{-1} H P.$$

The implementation uses the corresponding discrete prediction and Joseph-form correction.

Chapter 9. Discrete-Time Prediction

For sample interval h , a linear time-invariant SDE is discretized as

$$\begin{aligned}x_{k+1|k} &= \Phi(h) x_k, \\P_{k+1|k} &= \Phi(h) P_k \Phi(h)^T + Q_d(h), \\ \Phi(h) &= \exp(F h), \\ Q_d(h) &= \int_0^h \exp(F t) L Q_c L^T \exp(F^T t) dt.\end{aligned}$$

The Ocean IMU implementation computes the important OU-driven blocks analytically. This avoids numerical instability and avoids repeatedly using a general-purpose matrix exponential on small embedded targets.

9.1. OU Axis Transition

For one axis with state $[v, p, S, a^T]$, define

$$\begin{aligned}x &= h / \tau, \\ \alpha &= e^{-x}.\end{aligned}$$

The per-axis transition has the form

$$\Phi_{\text{axis}} = \begin{bmatrix} 1, & 0, & 0, & \phi_{va}; \\ h, & 1, & 0, & \phi_{pa}; \\ 1/2 h^2, & h, & 1, & \phi_{sa}; \\ 0, & 0, & 0, & \alpha \end{bmatrix}.$$

The coupling terms are

$$\begin{aligned}\phi_{va} &= \tau (1 - \alpha), \\ \phi_{pa} &= \tau^2 (x + \alpha - 1), \\ \phi_{sa} &= \tau^3 (1/2 x^2 - x - \alpha + 1).\end{aligned}$$

The full 3D OU-chain transition is the block diagonal stacking of three such axis transitions. The covariance Q_d is computed from an exact coefficient matrix for normal x and from a Maclaurin branch for small x , followed by symmetrization and positive-semidefinite cleanup.

Chapter 10. Rank-3 Measurement Updates

The accelerometer, magnetometer, and pseudo-measurement channels are all rank-3 updates. For a measurement

$$z = h(x) + nu,$$

the EKF update is

$$r = z - h(x^{\wedge-}),$$

$$S_{\text{innov}} = H P^{\wedge-} H^T + R,$$

$$K = P^{\wedge-} H^T S_{\text{innov}}^{-1},$$

$$x^{\wedge+} = x^{\wedge-} + K r.$$

The covariance update should use the Joseph form:

$$P^{\wedge+} = (I - K H) P^{\wedge-} (I - K H)^T + K R K^T.$$

After each correction, the attitude-error subvector is injected back into the nominal quaternion:

$$q \leftarrow \exp(1/2 \text{ delta theta}) \otimes q,$$

$$\text{delta theta} \leftarrow 0.$$

The rank-3 structure is important for embedded use. The expensive inversion is only a 3 by 3 solve for each triad update.

Chapter 11. Initialization and Reference Alignment

Startup should be staged. The filter should not immediately trust yaw, acceleration bias, displacement, or drift-control states.

A practical sequence is:

1. Estimate gravity direction from accelerometer data and initialize roll/pitch.
2. Leave yaw arbitrary until a stable magnetic reference is available.
3. Run an early attitude-only or reduced-state phase while the frequency tracker gathers statistics.
4. Enable the full linear wave block after the tuner has usable estimates of frequency and acceleration variance.
5. Delay magnetometer yaw correction until tilt is stable and magnetic samples have been accumulated in the tilt frame.
6. Enter live operation with online adaptation enabled.

This avoids seeding displacement, bias, and pseudo-measurement states with startup transients.

Chapter 12. Frequency Tracking

Adaptive tuning requires an online estimate of the dominant wave frequency. The signal is derived from vertical inertial acceleration, typically as an up-positive scalar:

```
a_up = -a_z,inert.
```

Three embedded-friendly trackers are supported in Ocean IMU:

Tracker	Role
KalmANF adaptive notch	Models the signal with a second-order notch and updates the notch parameter with a scalar Kalman step.
Aranovski observer	Uses a nonlinear sinusoidal frequency observer with a filtered input and adaptive auxiliary state.
PLL	Uses a band-passed signal, I/Q demodulation, and a bounded PI phase-locked loop.

The raw frequency is clamped and smoothed to produce `f_tune`, which drives the sea-state tuner and the horizontal wave-direction estimator.

Chapter 13. Variance-Informed Online Adaptation

The tuner adapts three quantities:

- OU correlation time τ ;
- stationary acceleration scale σ_a ;
- integral pseudo-measurement level R_S .

The variance estimate is tied to a fixed number of wave periods so that the adaptation horizon scales naturally between chop and swell:

```
T_eff = 1 / f_tune,  
  
tau_var = clip(K_periods T_eff, tau_var,min, tau_var,max),  
  
alpha_var = 1 - e^(-Delta t / tau_var).
```

After subtracting a configured acceleration noise floor,

```
sigma_wave^2 = max(0, sigma_tot^2 - sigma_nf^2),
```

the target parameters are

```
tau_star = c_tau / (2 f_tune),  
  
sigma_a,star = c_sigma sqrt(max(sigma_wave^2, epsilon)),  
  
R_S,star = c_R sigma_a,star tau_star^3.
```

The applied values are exponentially smoothed. τ and σ_a use a fixed adaptation time constant. R_S is smoothed with a time constant proportional to the current wave time scale, so it responds faster in short-period seas and slower in long-period swell.

13.1. Why R_S Scales as $\sigma_a \tau^3$

The pseudo-measurement is not arbitrary. The triple-integral state S is the result of integrating acceleration three times:

$$d v / d t = a,$$

$$d p / d t = v,$$

$$d S / d t = p.$$

In the frequency domain, the transfer from acceleration to S has magnitude squared proportional to $1 / \omega^6$. For a broad class of wave spectra, after separating spectrum scale from spectrum shape, the natural spread of S follows

$$\sigma_S \sim c_{\text{spec}} \sigma_a \tau^3.$$

Therefore a practical sea-state-invariant pseudo-measurement law is

$$R_S = k_i \sigma_a \tau^3.$$

This makes a calm sea and a steep sea produce comparable normalized pseudo-measurement innovations. The filter does not become overly stiff in large waves or overly loose in calm conditions purely because the acceleration level changed.

13.2. Anisotropy

The implementation may use different horizontal and vertical settings:

$$\Sigma_{aw}^{(1/2)} = \text{diag}(S_{\text{sigma}} \sigma_a, S_{\text{sigma}} \sigma_a, \sigma_a),$$

$$R_S = \text{diag}(\rho_{xy} R_S, \rho_{xy} R_S, R_S).$$

This allows stronger or weaker horizontal regularization without changing the vertical heave behavior.

Chapter 14. Horizontal Wave Direction

The filter estimates a slowly varying horizontal acceleration amplitude vector A_k . Let

$$z_k = [a_x(k), a_y(k)]^T.$$

For a dominant narrow-band wave with phase ϕ_k , the simplified observation model is

$$z_k \approx \cos(\phi_k) A_k + \eta_k.$$

This is the online Kalman-filter form of a cosine-projection least-squares fit:

$$A_{LS} = (\sum_k \cos(\phi_k) z_k) / (\sum_k \cos(\phi_k)^2).$$

The 2D direction filter uses

$$\begin{aligned} A_{(k+1)} &= A_k + w_k, \\ H_k &= \cos(\phi_k) I_2, \\ z_k &= H_k A_k + \eta_k. \end{aligned}$$

When $\text{abs}(\cos(\phi_k))$ is too small, the update is skipped because the observation carries little information near horizontal zero crossings.

The wave line angle is

$$\begin{aligned} d_{\text{hat}} &= A_k / ||A_k||, \\ \theta &= \text{wrap}_{[0,180)}(\text{atan2}(d_y, d_x) \ 180 / \pi). \end{aligned}$$

This gives direction modulo 180 deg. A separate vertical-horizontal phase discriminator can choose the travel sense along that line, producing a directed azimuth when the sign estimate is stable.

Chapter 15. Validation Summary

Ocean IMU validates the filter on synthetic directional sea states with controlled ground truth. The simulations use 20 minute runs at 200 Hz, with IMU and magnetometer error models representative of low-cost MEMS hardware.

15.1. Wave Scenarios

Scenario	Model	H _s m	T _p s	Directional spread
W1	JONSWAP, PM-Stokes	0.27	3.0	+30 deg / cos-spread
W2	JONSWAP, PM-Stokes	1.50	5.7	-30 deg / cos-spread
W3	JONSWAP, PM-Stokes	4.00	8.5	+30 deg / cos-spread
W4	JONSWAP, PM-Stokes	8.50	11.4	-30 deg / cos-spread

15.2. Simulation Results

The table below summarizes the last 60 seconds RMS window from the Ocean IMU validation run. **R/P/Y** are roll, pitch, and yaw RMS errors in degrees. **T/A/U** means Toward, Away, and Uncertain percentage of samples for the sign discriminator.

Case	H _s	Z RMS m	Z % H _s	R deg	P deg	Y deg	mean theta deg	T/A/U %
JONSWAP	0.27	0.026	9.7	0.42	0.30	1.11	29.2	99.5/0.0/0.5
JONSWAP	1.50	0.085	5.7	0.45	0.65	0.34	-32.7	100.0/0.0/0.1
JONSWAP	4.00	0.297	7.4	0.60	1.36	1.59	31.5	99.3/0.7/0.0
JONSWAP	8.50	0.453	5.3	1.37	0.54	1.06	-30.8	99.0/0.6/0.4
PM+Stokes	0.27	0.020	7.6	0.37	0.12	1.94	37.3	99.7/0.0/0.3
PM+Stokes	1.50	0.063	4.2	0.18	0.17	2.10	-34.5	99.7/0.0/0.3
PM+Stokes	4.00	0.262	6.6	0.19	0.39	2.10	26.4	100.0/0.0/0.0
PM+Stokes	8.50	0.537	6.3	1.00	0.56	2.21	-19.1	99.8/0.0/0.2

The simulation results show drift-suppressed vertical displacement while retaining realistic wave-band motion. The model also keeps roll, pitch, yaw, and wave-direction estimates bounded under sensor noise, residual bias, and multicomponent directional seas.

Chapter 16. Embedded Hardware Platform

The implementation has also been exercised on real embedded hardware using the M5Stack AtomS3R ESP32-S3 module. The purpose of this validation is to confirm real-time behavior and implementation feasibility:

- fixed-rate IMU sampling;
- bounded memory use;
- calibrated sensor triads;
- quaternion propagation from gyroscope data;
- rank-3 accelerometer updates at IMU rate;
- delayed magnetometer yaw correction when available;
- adaptive frequency and sea-state tuning from the vertical acceleration-like signal.

Quantitative wave-kinematics accuracy is evaluated in simulation, where exact displacement, velocity, acceleration, and attitude ground truth are available. Real hardware validates the embedded execution path and sensor-processing pipeline.

Chapter 17. Practical Implementation Notes

17.1. Prefer a Unified Filter Over a Pipeline

The mature method does not first estimate attitude, then separately integrate acceleration, then separately estimate wave spectra. It keeps attitude, acceleration, velocity, displacement, and drift control in one covariance model. This is important because tilt error and acceleration error are coupled by gravity.

17.2. Do Not Treat the Pseudo-Measurement as a Sensor

The $S = 0$ pseudo-measurement is an artificial stabilizer. Its covariance is a tuning parameter. It should be adapted from frequency and acceleration variance, not chosen as a fixed physical sensor noise.

17.3. Use Calibrated Axes

All equations assume that accelerometer, gyro, and magnetometer readings have already been calibrated and mapped into the expected body-frame convention. Hard-iron, soft-iron, scale, axis alignment, and gyro/accelerometer bias calibration should be handled before the estimator input stage.

17.4. Correct Lever Arm When Needed

An IMU mounted away from the vessel center of gravity will see angular-acceleration and centripetal terms. For small boats and low-cost modules this can be a large source of error. Use the lever-arm correction when the installation geometry is known.

17.5. Delay Magnetometer Trust

Magnetometer data are useful for yaw, but they are easily corrupted by local magnetic disturbances. The estimator should initialize tilt first, accumulate a stable magnetic reference, and only then enable regular yaw updates.

17.6. Barometer Is Not the Primary Wave Sensor

A barometer can be useful for weather and slow altitude trends, but it should not be treated as a direct wave-elevation sensor on a floating platform. The wave surface is not a static height field measured by air pressure in the way hiking or stair-climbing algorithms assume.

Chapter 18. Relationship to the Old Draft

The old document contained useful early notes about why double integration drifts and why pressure is not a reliable wave-height source on the wave surface. Those points remain valid. However, the trochoidal reconstruction and Doppler-based approach is no longer the preferred core algorithm.

The current preferred algorithm is:

1. calibrated IMU and optional magnetometer input;
2. quaternion MEKF for attitude and gyro bias;
3. OU-driven 3D kinematic chain for wave acceleration, velocity, displacement, and integral displacement;
4. adaptive pseudo-measurement drift control;
5. online frequency tracking and wave-direction estimation;
6. simulation validation against directional JONSWAP and PM-Stokes sea states;
7. embedded execution on ESP32-S3-class hardware.

Chapter 19. References

- Bareboat Necessities Ocean IMU project: <https://github.com/bareboat-necessities/ocean-imu>
- M. S. Grushinskiy, *OU-Driven Quaternion MEKF for Marine INS and Wave-State Estimation*, draft in [ocean-imu/doc/kalman_ou_iii](#).
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